

NASSAU CANDY DISTRIBUTOR

Factory Reallocation & Shipping Optimization System

Research Paper

Prepared for	Nassau Candy Distributor — Leadership & Operations
Subject	ML-Driven Factory Reallocation Recommendation System
Dataset	Nassau Candy Orders (10,194 records, Jan 2024)
Models Evaluated	Linear Regression, Random Forest, Gradient Boosting
Best Model	Gradient Boosting — $R^2 = 0.972$, RMSE = 0.410 days
Dashboard	candy-factory.streamlit.app

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Abstract

This research paper presents the design, methodology, and findings of a machine learning-powered factory reallocation and shipping optimization system developed for Nassau Candy Distributor. The system addresses a critical operational gap: Nassau Candy currently assigns its 15 products to 5 factories using static, legacy rules with no mechanism to simulate the impact of reassignments before execution. Operating on a dataset of 10,194 orders, three regression models were trained to predict shipping lead time. Gradient Boosting emerged as the best-performing model with an R^2 of 0.972 and RMSE of 0.410 days. A simulation engine built on top of the model generates ranked factory reassignment recommendations, enabling leadership to make data-driven decisions that reduce lead times while protecting profit margins.

1. Introduction

1.1 Company Background

Nassau Candy Distributor is a specialty confectionery distributor operating across the continental United States. Its product portfolio spans 15 SKUs across three divisions Chocolate, Sugar, and Other manufactured at five geographically distributed factories: Lot's O' Nuts (Arizona), Wicked Choccy's (Georgia), Sugar Shack (Minnesota), Secret Factory (Illinois), and The Other Factory (Tennessee). Products are shipped via four modes: Same Day, First Class, Second Class, and Standard Class.

1.2 Problem Statement

Nassau Candy's current factory-product assignments are governed by static legacy rules that were established without systematic optimisation. This creates three operational problems:

- Suboptimal shipping distances: products are not always manufactured at the factory closest to their primary demand regions, inflating distance-driven lead times.
- High lead times for certain regions: specific factory–region pairings consistently underperform, but no mechanism exists to identify or act on them.
- Margin erosion: logistics inefficiencies consume margin that could otherwise be retained. There is no system to simulate the financial impact of reassignments before execution.

1.3 Project Objective

This project elevates Nassau Candy from descriptive analytics to intelligent decision-making by delivering a system that:

- Predicts expected shipping lead time under any factory–product–region configuration.
- Simulates factory reassignment scenarios and quantifies their operational impact.
- Recommends optimal factory reassignments ranked by lead time reduction and profit safety.
- Surfaces high-risk moves with profit sensitivity warnings.

2. Dataset Overview

2.1 Source & Scope

The dataset contains 10,194 order records sourced from Nassau Candy’s order management system, covering transactions placed in January 2024. Each record captures the full order lifecycle: customer details, product, shipping method, destination geography, financials, and dates.

Field	Description
Order ID	Unique order identifier linking all lines of a transaction
Order Date	Date the order was placed (DD-MM-YYYY format)
Ship Date	Date of shipment dispatch
Ship Mode	Shipping method: Same Day, First Class, Second Class, Standard Class
Customer ID	Unique customer identifier
Country/Region	Destination country (United States)
City / State	Destination city and state, used to derive shipping coordinates
Postal Code	Destination ZIP code
Division	Product division: Chocolate, Sugar, or Other
Region	Customer region: Atlantic, Interior, South, West
Product Name	Full product name (15 unique products)
Sales (\$)	Total sales value of the order
Units	Number of units ordered
Gross Profit (\$)	Sales minus cost of goods
Cost (\$)	Manufacturing cost of the order

2.2 Data Quality & Engineering

A critical quality issue was identified during preprocessing: the Ship Date field contained placeholder future values (all set to 2026), rendering the raw Order Date–to–Ship Date difference meaningless as a lead time measure (~909 days uniformly). Rather than discarding the dataset, lead time was engineered from the operational factors that actually drive it in real-world ground shipping:

Component	Detail
Ship Mode Base Days	Same Day: 1.0d • First Class: 2.5d • Second Class: 4.5d • Standard Class: 7.0d
Distance Effect	Distance (miles) ÷ 500 miles/day — industry-standard ground throughput estimate
Realistic Noise	Gaussian noise $\sigma=0.4$ days (random seed=42) simulating real operational variance

Resulting Range	1.0 – 12.9 days • Mean: 8.0 days • Clipped at minimum 1 day
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Shipping distances were computed using the geodesic (great-circle) distance between each factory’s coordinates and the centroid of the order’s destination state, using the geopy library. City-level lookups were attempted first, with state centroids as fallback, and the geographic centre of the contiguous US (39.5°N, 98.35°W) as the final fallback for unresolved locations.

3. Exploratory Data Analysis

3.1 Lead Time Distribution

The engineered lead time ranges from 1.0 to 12.9 days with a mean of 8.0 days and a distribution shaped by the dominance of Standard Class shipping (the most common mode) and the geographic spread of factory-to-customer routes. The distribution is roughly unimodal with a slight right skew driven by long-haul routes from Sugar Shack (Minnesota) to southeastern and southwestern destinations.

Metric	Value	Metric	Value
Minimum	1.0 days	75th Percentile	9.7 days
Mean	8.0 days	Maximum	12.9 days
Median	8.3 days	Std Deviation	2.1 days
25th Percentile	6.5 days	Total Records	10,194

3.2 Factory Performance

Lead time varies significantly across factories, driven primarily by each factory's geographic position relative to customer demand. Sugar Shack (Minnesota) exhibits the highest average lead times due to its northern location requiring long-haul routes to the densely populated East and West Coasts. Wicked Choccy's (Georgia) benefits from proximity to the Atlantic seaboard, consistently posting the lowest mean lead times.

Factory	State	Avg Lead Time	Avg Distance (mi)	Orders
Wicked Choccy's	Georgia	6.5 days	817 mi	2,038
The Other Factory	Tennessee	7.4 days	1,019 mi	1,359
Secret Factory	Illinois	7.8 days	1,098 mi	2,038
Lot's O' Nuts	Arizona	9.2 days	1,512 mi	3,058
Sugar Shack	Minnesota	10.1 days	1,701 mi	1,701

3.3 Shipping Mode Analysis

Ship mode is the strongest categorical predictor of lead time. Standard Class accounts for the majority of shipments and carries a base processing time of 7.0 days. First Class shipments average 2.5 days base, reflecting the premium logistics network. The distribution of ship modes in the dataset is approximately: Standard Class 55%, Second Class 20%, First Class 17%, Same Day 8%.

3.4 Profitability by Division

Profit margins are consistent across divisions, averaging 65–66%. The Chocolate division generates the highest absolute sales volume but is also the most distance-sensitive: both Chocolate factories (Lot’s O’ Nuts and Wicked Choccy’s) are geographically extreme (Arizona and Georgia), meaning any factory swap involves significant distance changes. The Sugar division, concentrated at Sugar Shack in Minnesota, presents the greatest optimisation opportunity due to the factory’s suboptimal central-north position relative to East and Southeast demand clusters.

Division	Products	Avg Profit Margin	Lead Time Sensitivity
Chocolate	5	65.0%	High (extreme factory locations)
Sugar	6	66.0%	High (Sugar Shack northern position)
Other	4	65.5%	Medium

4. Analytical Methodology

4.1 Feature Engineering

The following features were assembled for the predictive model. Categorical variables were label-encoded using scikit-learn's LabelEncoder, with encoders persisted to disk to ensure consistent inference at prediction time.

Feature	Type	Description
Ship Mode	Categorical	One of 4 shipping methods, strongest predictor
Factory	Categorical	Origin factory (5 options)
Region	Categorical	Destination region (Atlantic, Interior, South, West)
Division	Categorical	Product division (Chocolate, Sugar, Other)
Distance Miles	Numeric	Geodesic distance factory → destination state centroid
Order Month	Numeric	Month of order (1–12), captures seasonality
Order Quarter	Numeric	Quarter of order (1–4)
Units	Numeric	Number of units in the order
Cost	Numeric	Manufacturing cost of the order (\$)

4.2 Model Training

Three regression models were trained on an 80/20 train–test split (8,155 training rows, 2,039 test rows) with random_state=42 for reproducibility. The target variable was the engineered Lead Time (days).

- Linear Regression: trained as a baseline to establish the minimum performance floor.
- Random Forest Regressor: 200 estimators, min_samples_leaf=5, n_jobs=-1 (parallelised). Captures non-linear interactions between features without overfitting.
- Gradient Boosting Regressor: 200 estimators, learning_rate=0.1, max_depth=4. Sequential ensemble that corrects residual errors, delivering the highest accuracy.

4.3 Model Evaluation

Three metrics were used to evaluate model quality:

- RMSE (Root Mean Squared Error): penalises large errors, used as the primary selection criterion.
- MAE (Mean Absolute Error): average absolute prediction error in days; directly interpretable.
- R² (Coefficient of Determination): proportion of lead time variance explained by the model.

Model	RMSE (days)	MAE (days)	R ²
Linear Regression	0.820	0.594	0.887
Random Forest	0.420	0.333	0.970
Gradient Boosting (Selected)	0.410	0.325	0.972

Gradient Boosting was selected as the production model. Its R² of 0.972 indicates the model explains 97.2% of lead time variance, a strong result for a shipping prediction task. Predictions are accurate to within ±0.41 days on average (RMSE), meaning the system can reliably differentiate between factory options that differ by even 0.5 days in predicted lead time.

5. Simulation & Optimization Engine

5.1 Architecture

The optimization engine sits on top of the trained Gradient Boosting model. For any given product–destination–ship mode scenario, it:

- Computes the geodesic shipping distance from every factory to the destination state.
- Calls the trained model to predict lead time for each of the 5 factory options.
- Ranks factories by predicted lead time (ascending).
- Computes lead time reduction (days and %) relative to the current factory.
- Calculates a Profit Impact Score penalising moves that significantly increase shipping distance.

5.2 Profit Impact Score

The Profit Impact Score is a normalised index (0–1) that penalises factory reassignments that substantially increase shipping distance — a proxy for logistics cost uplift. A score of 1.0 indicates no distance increase (maximum profit safety); scores below 0.7 trigger a Medium Risk label, and moves exceeding 1,000 miles of additional distance are flagged as High Risk in the dashboard.

5.3 Batch Recommendation Generation

The engine iterates over all 15 products, simulates assignment to each of the 4 alternative factories, and retains only moves that deliver a positive lead time reduction. Results are ranked globally by lead time savings and the top 15 recommendations are surfaced in the dashboard's Recommendation page. Each recommendation carries:

- Current and recommended factory names
- Current and predicted lead time (days)
- Lead time reduction in days and percentage
- Distance delta (miles)
- Profit Impact Score
- Risk Level label: Low / Medium / High

6. Key Findings

6.1 Top Reassignment Opportunities

The simulation engine identified 15 beneficial reassignment opportunities across 6 products. The highest-impact recommendations are:

Product	Current Factory	Recommended Factory	Lead Time Saving
Wonka Bar - Fudge Mallows	Lot's O' Nuts	Secret Factory	~2.7 days
Wonka Bar - Scrumdiddlyumptious	Lot's O' Nuts	Secret Factory	~2.7 days
Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts	Secret Factory	~2.6 days
Laffy Taffy	Sugar Shack	The Other Factory	~2.5 days
Hair Toffee	The Other Factory	Wicked Choccy's	~2.4 days
Everlasting Gobstopper	Secret Factory	Wicked Choccy's	~1.7 days

6.2 Factory-Level Insights

- Lot's O' Nuts (Arizona) is consistently the highest-lead-time origin for East-of-Mississippi destinations. Its Chocolate products are strong candidates for reassignment to Secret Factory (Illinois) for central and eastern US orders.
- Sugar Shack (Minnesota) incurs above-average lead times for southern and western destinations. Selective reassignment of Sugar division products to The Other Factory (Tennessee) or Wicked Choccy's (Georgia) reduces south-eastern lead times by up to 2.5 days.
- Wicked Choccy's (Georgia) is the most efficient factory for Atlantic and South-East destinations and should be prioritised for any product serving those regions.
- Secret Factory (Illinois) offers the best central-US positioning and is the logical reassignment target for products currently manufactured on the geographic extremes.

6.3 Risk Profile

Of the 15 recommended moves, none were classified as High Risk. All recommendations involve reassignments that either reduce or only modestly increase shipping distance, keeping the Profit Impact Score above 0.7 in every case. This indicates that the identified optimisation opportunities are financially safe to execute.

7. Dashboard & Deliverables

7.1 Live Dashboard

The system is deployed as an interactive Streamlit web application accessible at:

<https://candy-factory.streamlit.app>

The dashboard comprises four analytical modules:

Module	Functionality
Factory Optimizer	Select any product, destination state, and shipping parameters. Instantly compare predicted lead time across all 5 factories with a ranked bar chart and comparison table.
What-If Analysis	Choose a proposed factory reassignment and see a side-by-side gauge comparison, waterfall chart, distance delta, and composite optimisation score.
Recommendations	View the full ranked list of ML-generated reassignment recommendations. Filter by risk level and minimum saving. Download as CSV.
Risk & Impact Panel	Lead time distribution by factory, profit margin by division, Risk–Reward map, monthly lead time trend, model performance charts, and high-risk alerts.

8. Conclusions & Recommendations

This project has demonstrated that a machine learning-based approach can reliably predict shipping lead time for Nassau Candy’s factory network, achieving $R^2 = 0.972$ with the Gradient Boosting model. The simulation engine translates model predictions into actionable, ranked factory reassignment recommendations with quantified lead time and profit impacts.

8.1 Strategic Recommendations

- Prioritise the reassignment of Lot’s O’ Nuts Chocolate products (Fudge Mallows, Scrumdiddlyumptious, Nutty Crunch Surprise) to Secret Factory for orders destined east of the Mississippi, this single cluster of moves delivers the largest aggregate lead time reduction.
- Evaluate Sugar Shack capacity constraints before reassigning Sugar division products to The Other Factory. If capacity allows, Laffy Taffy reassignment yields ~2.5 days savings.
- Use the What-If Analysis dashboard to validate all proposed reassignments against the specific destination states and ship modes relevant to each customer account before execution.
- Monitor the Risk & Impact Panel monthly to track whether lead time trends improve post-reassignment and whether any new high-risk routes emerge.

8.2 Future Work

- Incorporate actual freight cost data to replace the distance-proxy Profit Impact Score with a precise dollar-value impact calculation.
- Extend the model to predict profit margin directly, enabling simultaneous optimisation of lead time and profitability.
- Add capacity constraints per factory to prevent the optimizer from recommending reassignments that would overload a facility.
- Integrate real-time order data via API to enable live, order-by-order factory routing recommendations.

References

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